Elementary Statistics for Social Sciences II

208272 (Part 1)

Asst. Prof. Dr. Parichart Pattarapanitchai

2026-10-14

Table of contents

Preface 4
- How to use this book . . . 4
- Notation for critical values . . . 5
- The five steps of a hypothesis test . . . 5
- Writing the conclusion . . . 5

1 Basic Principles of Statistical Inference 7
- Learning Objectives . . . 7
- 1.1 Parameters, Statistics and Sampling Distributions . . . 7
- 1.2 Standard Error and the Central Limit Theorem . . . 9
- 1.3 The Idea of a Confidence Interval . . . 12
- 1.4 The Logic of Hypothesis Testing . . . 14
- 1.5 Type I and Type II Errors . . . 16
- 1.6 Reading Statistical Tables . . . 16
- Formula Summary . . . 18
- Exercises . . . 19
- Solutions . . . 20

2 One-Sample Inference 22
- 2.1 Mean, σ Known . . . 22
- 2.2 Mean, σ Unknown . . . 24
- 2.3 Proportion . . . 26
- 2.4 Variance . . . 28

3 Two-Sample Inference 31
- 3.1 Two Proportions . . . 31
- 3.2 P-values and R Output . . . 33
- 3.3 Paired Means . . . 36
- 3.4 Two Variances . . . 38
- 3.5 Two Independent Means . . . 41

4 Analysis of Variance 44
- 4.1 One-Way ANOVA: CRD . . . 44
- 4.2 Multiple Comparisons (Tukey) . . . 46
- 4.3 Randomized Complete Block Design (RCBD) . . . 49

Summary: Choosing a Method **52**
Which Method? A Flowchart 52
Mixed Problems 52
Writing a Full Conclusion 52

Preface

This book covers the first half of 208272 Elementary Statistics for Social Sciences II: the principles of statistical inference, inference about means, proportions and variances, and analysis of variance. All examples use economic contexts. Unless a source is given, the numbers in the examples are hypothetical and are used for illustration only.

How to use this book

Each section of Chapters 2–4 matches one class session and follows the same pattern:

1. **Learning objectives**
2. **Concept**: why we need this method
3. **Formulas and conditions**
4. **Worked example**: a full solution
5. **Your Turn**: a similar problem with space to write your own solution
6. **Reading R output** (from Section 3.2 onward)
7. **Formula summary**
8. **Exercises**
9. **Solutions** to Your Turn and the Exercises

In the PDF, the Your Turn problems have blank space for writing, and the solutions are at the end of each section. On the web, click a solution box to open it. Try the problem first.

Session	Section
1	Chapter 1 Basic Principles of Statistical Inference
2	2.1 Mean, σ known
3	2.2 Mean, σ unknown
4	2.3 Proportion
5	2.4 Variance
6	3.1 Two proportions
7	3.2 P-values and R output; 3.3 Paired means
8	3.4 Two variances
9	3.5 Two independent means
10	4.1 One-way ANOVA: CRD
11	4.2 Multiple comparisons (Tukey); 4.3 RCBD
12	Summary: Choosing a Method

Notation for critical values

The subscript of a critical value is the **area to its left**. This is the same convention as the R functions `qnorm()`, `qt()`, `qchisq()` and `qf()`.

Notation	Meaning	R
$z_{0.975}$	the z value with area 0.975 to its left	`qnorm(0.975)`
$t_{0.975,\,24}$	the t value with 24 df and area 0.975 to its left	`qt(0.975, 24)`
$\chi^2_{0.025,\,10}$	the χ^2 value with 10 df and area 0.025 to its left	`qchisq(0.025, 10)`
$F_{0.975,\,5,\,8}$	the F value with (5, 8) df and area 0.975 to its left	`qf(0.975, 5, 8)`

The statistical tables for this course give the same left-tail areas:

Distribution	Table	Left-tail areas in the table
Standard normal (z)	Table 4	z from 0.00 to 3.49
t	Table 5	0.60, 0.70, 0.80, 0.90, 0.95, 0.975, 0.99, 0.995, 0.999, 0.9995
Chi-square (χ^2)	Table 6	0.0005 to 0.9995
F	Table 7	0.90, 0.95, 0.975, 0.99, 0.995

A value calculated from the sample has the subscript "cal", for example z_{cal}, t_{cal}, χ^2_{cal} and F_{cal}.

The five steps of a hypothesis test

1. **Hypotheses**: state H_0 and H_1.
2. **Level of significance**: state α.
3. **Test statistic**: write the formula, substitute the values, and give the result.
4. **Critical value and critical region**.
5. **Decision and interpretation**.

Writing the conclusion

Decision using the critical region (Chapters 1–2 and Section 3.1)

t_{cal} = … [falls / does not fall] in the critical region (t_{cal} [<, >] …), so we [reject / accept] H_0.

Decision using the p-value (from Section 3.2 onward)

p-value = … [<, ≥] $\alpha = 0.05$, so we [reject / accept] H_0.

Interpretation

That is, … at the 0.05 level of significance.

- If we reject H_0, write H_1 as a sentence.
- If we accept H_0, write H_0 as a sentence.
- For a two-sided test that rejects H_0, also state the direction, using the sample statistic.

Confidence interval

We are 95% confident that … is between … and ….

1 Basic Principles of Statistical Inference

Learning Objectives

After this chapter you should be able to:

1. Distinguish a **parameter** from a **statistic**.
2. Describe the sampling distribution of $\bar{x}$ and of $\hat{p}$: its mean, its standard error and its shape.
3. Use the Central Limit Theorem to find probabilities about $\bar{x}$ and $\hat{p}$.
4. Explain what a 95% confidence interval means.
5. Write the null and alternative hypotheses for a research question, and tell a one-sided test from a two-sided test.
6. Find critical values and the critical region from the z table (Table 4).
7. Explain Type I and Type II errors in an economic setting.

> Why Does This Matter?
>
> Economists rarely see a whole population. The Ministry of Labour cannot ask every worker in Chiang Mai about their wage, and the Bank of Thailand cannot survey every household about its debt. Instead we take a **sample** and use it to say something about the **population**.
> Two samples from the same population give different answers. Statistical inference tells us how much a sample result can vary, and so how far we can trust it. This chapter builds the ideas used in every later chapter.

1.1 Parameters, Statistics and Sampling Distributions

A **parameter** describes a population. It is a fixed number, but usually unknown. A **statistic** describes a sample. We can calculate it, but it changes from sample to sample.

	Population (parameter)	Sample (statistic)
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Variance	σ^2	s^2
Proportion	p	$\hat{p}$

	Population (parameter)	Sample (statistic)
Size	N	n

The **sampling distribution** of a statistic is the distribution of its values over all possible samples of the same size n.

Example. Figure 1.1 uses a hypothetical population of 100,000 households. Monthly household income is skewed to the right: most households earn a moderate amount and a few earn a great deal. The population mean is μ = 27,054 baht and the standard deviation is σ = 19,543 baht. The other panels show the means of 5,000 random samples of size 5, 30 and 100.

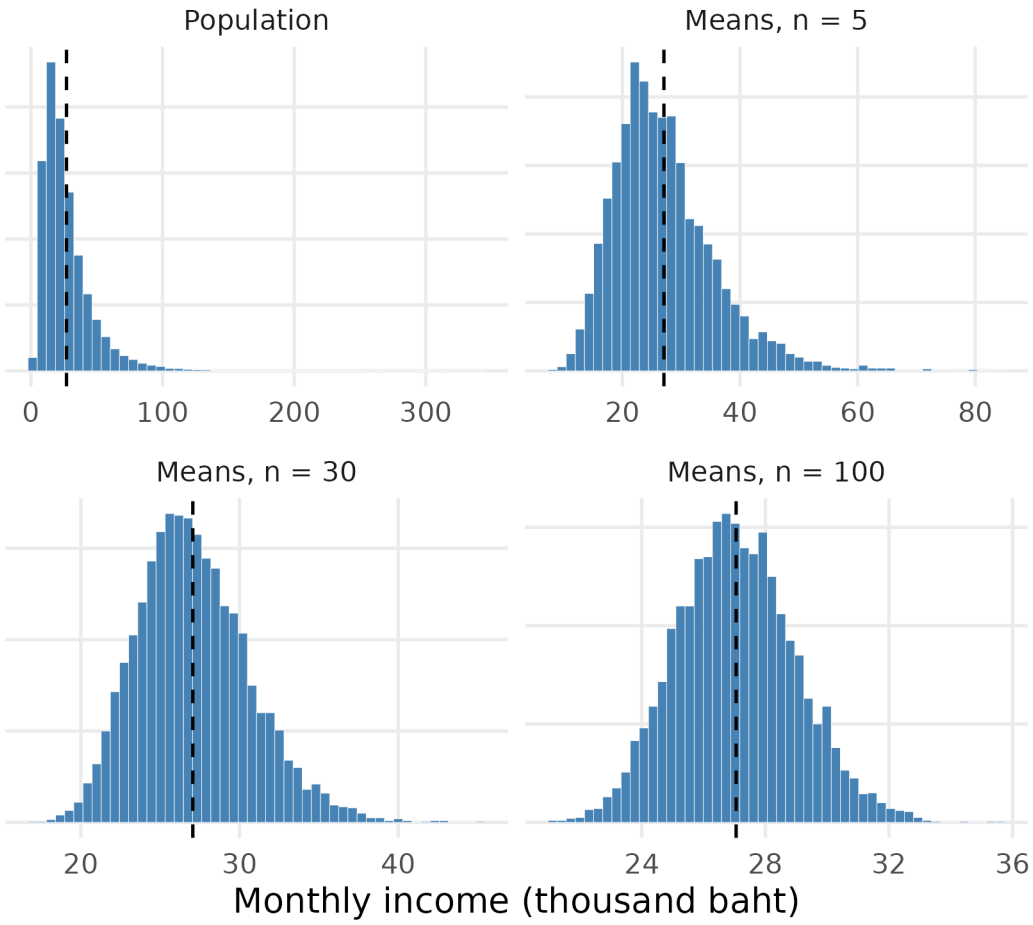


Figure 1.1: Household income in a hypothetical population (top left) and the sampling distribution of the sample mean for n = 5, 30 and 100

Three things happen as n grows:

1. The sample means are centred on the population mean μ (dashed line).
2. The spread of the sample means gets **smaller**.

3. The shape gets closer to a **normal** curve, even though the population is skewed.

1.2 Standard Error and the Central Limit Theorem

The standard deviation of a sampling distribution is called the **standard error (SE)**. It measures how much a statistic varies from sample to sample.

Sampling distribution of $\bar{x}$

- Mean: $\mu_{\bar{x}} = \mu$
- Standard error: $\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}$
- Shape:
  - if the population is normal, $\bar{x}$ is normal for any n;
  - if not, $\bar{x}$ is **approximately normal** when n is large ($n \geq 30$). This is the **Central Limit Theorem (CLT)**.

So we can standardize $\bar{x}$ and use the z table:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Sampling distribution of $\hat{p}$

The sample proportion $\hat{p} = x/n$, where x is the number of "successes".

- Mean: $\mu_{\hat{p}} = p$
- Standard error: $\sigma_{\hat{p}} = \sqrt{\dfrac{p(1-p)}{n}}$
- Shape: approximately normal when $np \geq 5$ and $n(1-p) \geq 5$.

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Larger samples, smaller error

Because $\sqrt{n}$ is in the denominator, a sample four times as large halves the standard error.

Worked Example 1.1: daily wage

Suppose the daily wage of construction workers in Chiang Mai has mean $\mu = 380$ baht and standard deviation $\sigma = 60$ baht (hypothetical figures). A random sample of $n = 36$ workers is taken.

(a) Describe the sampling distribution of $\bar{x}$.
(b) Find the probability that the sample mean wage is more than 400 baht.

Solution

(a) Since $n = 36 \geq 30$, by the CLT $\bar{x}$ is approximately normal with

$$\mu_{\bar{x}} = 380, \qquad \sigma_{\bar{x}} = \frac{60}{\sqrt{36}} = 10 \text{ baht}.$$

(b)

$$z = \frac{400 - 380}{10} = 2.00$$

From Table 4, $P(Z \leq 2.00) = 0.9772$, so

$$P(\bar{x} > 400) = P(Z > 2.00) = 1 - 0.9772 = 0.0228.$$

About 2.3% of samples of 36 workers would have a mean wage above 400 baht.

Worked Example 1.2: household debt

Suppose 65% of households in a region have outstanding debt (hypothetical figure). A survey samples $n = 400$ households. Find the probability that less than 60% of the sampled households have debt.

Solution

Check the conditions: $np = 260 \geq 5$ and $n(1-p) = 140 \geq 5$, so $\hat{p}$ is approximately normal.

$$\sigma_{\hat{p}} = \sqrt{\frac{0.65(1 - 0.65)}{400}} = 0.0238$$

$$z = \frac{0.60 - 0.65}{0.0238} = -2.10$$

Table 4 gives only positive z. By symmetry,

$$P(\hat{p} < 0.60) = P(Z < -2.10) = P(Z > 2.10) = 1 - 0.9821 = 0.0179.$$

Your Turn

Your Turn 1.1

The monthly rent of rooms near a university has mean 4,500 baht and standard deviation 900 baht (hypothetical figures). A random sample of 81 rooms is taken. Find

(a) the standard error of $\bar{x}$, and
(b) the probability that the sample mean rent is less than 4,320 baht.

Your Turn 1.2

In a province, 40% of workers are in the informal sector (hypothetical figure). A labour force survey samples 600 workers. Find the probability that more than 45% of the sample work in the informal sector.

1.3 The Idea of a Confidence Interval

A single number such as $\bar{x}$ is a **point estimate**. It is almost never exactly equal to μ. A **confidence interval (CI)** gives a range of plausible values instead:

$$\text{point estimate} \pm (\text{critical value}) \times \text{SE}$$

The **confidence level** (for example 95%) describes the method, not one interval: if we took many samples and built a 95% CI from each, about 95% of the intervals would contain the true parameter.

In Figure 1.2, 37 of the 40 intervals contain $\mu = 380$ and 3 do not. For a single interval we write:

> We are 95% confident that the mean daily wage is between … and … baht.

> **A common mistake**
>
> Do not say "there is a 95% probability that μ is in this interval". Once the interval is calculated, μ is either in it or not. The 95% refers to how often the method works.

Chapters 2 and 3 give the formula of the CI for each parameter.

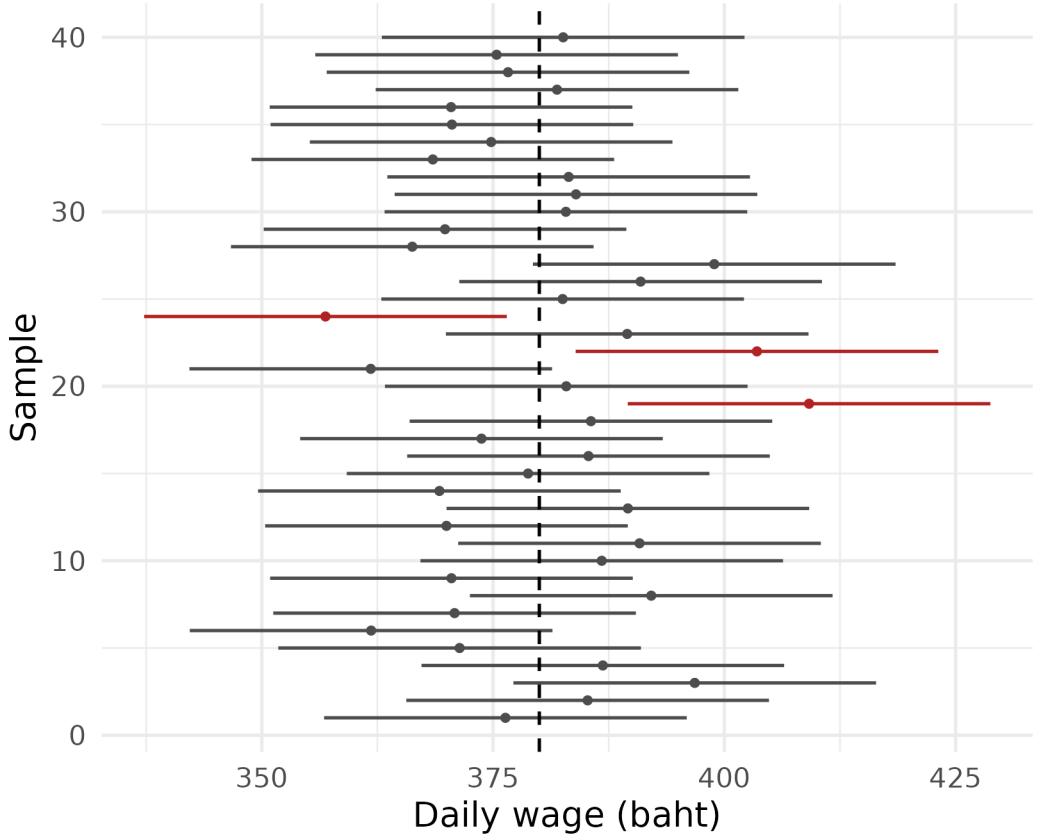


Figure 1.2: Forty 95% confidence intervals for the mean daily wage, each from a different sample of 36 workers (hypothetical population with mean 380 baht). Red intervals miss the true mean.

1.4 The Logic of Hypothesis Testing

A hypothesis test uses sample data to decide between two statements about a population parameter.

- The **null hypothesis** H_0 is the statement of "no change" or "no difference". It always contains the equals sign.
- The **alternative hypothesis** H_1 is what the researcher wants to show. It contains $\neq$, $<$ or $>$.

We assume H_0 is true and ask whether the sample result is unusual under that assumption. If it is, we reject H_0.

Example. The Ministry of Labour reports that the mean daily wage of construction workers is 380 baht. A labour union believes the mean is lower.

$$H_0 : \mu = 380 \qquad H_1 : \mu < 380$$

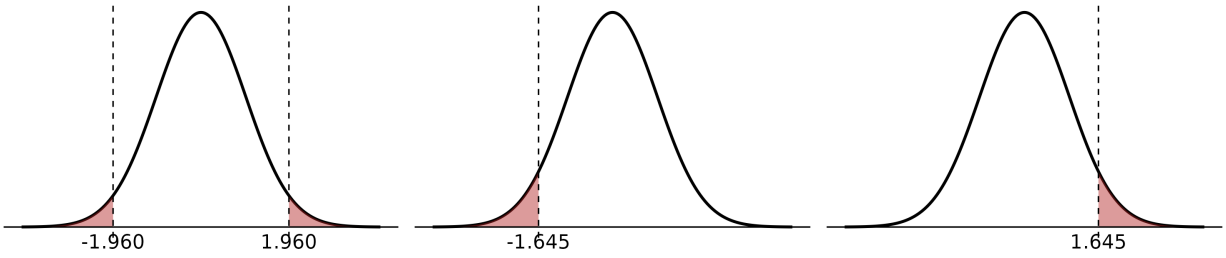
The direction in H_1 comes from the **research question**, never from the data.

Research question	H_1	Test	Critical region
Is it different from μ_0?	$\mu \neq \mu_0$	two-sided	both tails, area $\alpha/2$ each
Is it less than μ_0?	$\mu < \mu_0$	one-sided (left)	left tail, area α
Is it greater than μ_0?	$\mu > \mu_0$	one-sided (right)	right tail, area α

Test statistic, critical value and critical region

- The **test statistic** measures how far the sample result is from H_0, in standard errors. For example $z_{\text{cal}} = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$.
- The **level of significance** α is the probability of rejecting H_0 when H_0 is true. We usually use $\alpha = 0.05$.
- The **critical value** separates the values of the test statistic that lead to rejecting H_0 from those that do not.
- The **critical region** is the set of values that lead to rejecting H_0.

Decision rule. If z_{cal} falls in the critical region, we reject H_0. Otherwise we accept H_0.



(a) Two-sided (b) Left-sided (c) Right-sided

Figure 1.3: Critical regions for a z test at alpha = 0.05

Writing the hypotheses

Worked Example 1.3. Write H_0 and H_1.

(a) A price-monitoring agency checks whether the mean price of a 5-kg bag of rice has changed from 210 baht.
(b) An economist claims that more than 30% of households in Chiang Mai have used a government stimulus measure.
(c) A bank claims that the mean monthly loan payment of its customers is less than 8,000 baht.

Solution

(a) $H_0 : \mu = 210 \quad H_1 : \mu \neq 210$ (two-sided: "changed" has no direction)
(b) $H_0 : p = 0.30 \quad H_1 : p > 0.30$ (right-sided)
(c) $H_0 : \mu = 8000 \quad H_1 : \mu < 8000$ (left-sided)

Your Turn 1.3

Write H_0 and H_1, and say whether each test is two-sided, left-sided or right-sided.

(a) A researcher tests whether the mean household debt in a district is higher than 200,000 baht.
(b) An analyst tests whether the proportion of firms that raised prices last quarter is different from 0.50.
(c) A manager tests whether a new pricing strategy has reduced the mean number of days a product stays in stock below 14 days.

1.5 Type I and Type II Errors

The decision is based on a sample, so it can be wrong.

	H_0 is true	H_0 is false
Reject H_0	Type I error (probability α)	Correct decision
Accept H_0	Correct decision	Type II error (probability β)

Example. The government tests whether a stimulus measure increased mean household spending. H_0: the measure had no effect; H_1: the measure increased spending.

- **Type I error:** concluding that the measure worked when it did not. The government may continue an expensive policy that has no effect.
- **Type II error:** concluding that the measure did not work when it did. The government may stop a useful policy.

Choosing α

A smaller α makes a Type I error less likely but a Type II error more likely, for the same sample size. A larger sample reduces both.

1.6 Reading Statistical Tables

All the tables used in this course (Tables 4–7) give the **area to the left** of a value. The subscript in our notation is that area: $z_{0.975}$ is the value with area 0.975 to its left.

Two steps for every critical value

1. Draw the curve and **shade the area to the left** of the value you need.
2. Read the table at that left-tail area.

The z table (Table 4) gives $P(Z \leq z)$ for z from 0.00 to 3.49. The bottom row also lists the most common values:

Table 1.1: Common critical values of z (bottom row of Table 4)

Area to the left	z
0.900	1.282
0.950	1.645
0.975	1.960
0.990	2.326
0.995	2.576

Table 4 has only positive z. Because the normal curve is symmetric, $z_{0.05} = -z_{0.95}$ and $z_{0.025} = -z_{0.975}$.

Worked Example 1.4. Find the critical value(s) and critical region for a z test with $\alpha = 0.05$.

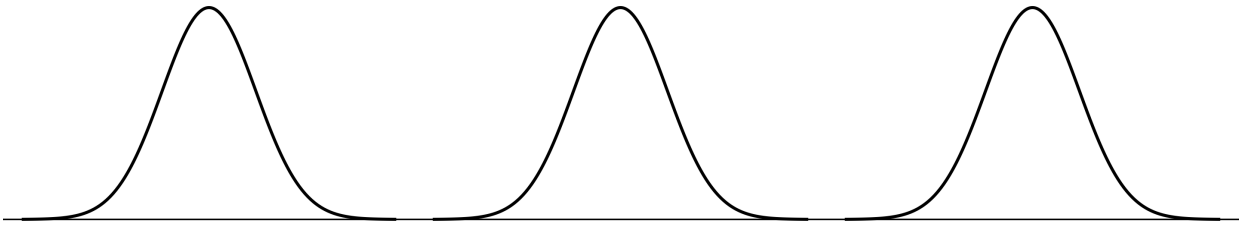
Solution

H_1	Area to the left	Critical value(s)	Critical region
$\neq$	0.025 and 0.975	$z_{0.025} = -1.960$, $z_{0.975} = 1.960$	$z_{\text{cal}} < -1.960$ or $z_{\text{cal}} > 1.960$
$<$	0.05	$z_{0.05} = -1.645$	$z_{\text{cal}} < -1.645$
$>$	0.95	$z_{0.95} = 1.645$	$z_{\text{cal}} > 1.645$

Your Turn 1.4

For each test, shade the area to the left on the curve first, then find the critical value(s) and write the critical region.

(a) $H_1 : \mu \neq \mu_0$, $\alpha = 0.01$
(b) $H_1 : \mu > \mu_0$, $\alpha = 0.01$
(c) $H_1 : p < p_0$, $\alpha = 0.10$



(a) (b) (c)

Figure 1.4: Shade the area to the left before reading Table 4

Formula Summary

Statistic	Mean	Standard error	Normal when	z
$\bar{x}$	μ	$\frac{\sigma}{\sqrt{n}}$	population normal, or $n \geq 30$	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$
$\hat{p}$	p	$\sqrt{\frac{p(1-p)}{n}}$	$np \geq 5$ and $n(1-p) \geq 5$	$\frac{\hat{p}-p}{\sqrt{p(1-p)/n}}$

H_1	Critical region (z test)
$\neq$	$z_{\text{cal}} < z_{\alpha/2}$ or $z_{\text{cal}} > z_{1-\alpha/2}$
$<$	$z_{\text{cal}} < z_{\alpha}$
$>$	$z_{\text{cal}} > z_{1-\alpha}$

Exercises

1.1 The price of a kilogram of pork at fresh markets has mean 45 baht and standard deviation 4 baht (hypothetical figures). A random sample of 64 markets is taken.

(a) Find the standard error of $\bar{x}$.
(b) Find $P(\bar{x} < 44.2)$.
(c) Find $P(44.2 < \bar{x} < 46)$.

1.2 Write H_0 and H_1.

(a) The Bank of Thailand tests whether mean headline inflation over the past year was above its 3% upper target.
(b) A researcher tests whether the proportion of graduates who find a job within six months differs from 0.70.
(c) A firm tests whether a discount campaign reduced the mean time to sell a product below 20 days.

1.3 Find the critical value(s) and the critical region of a z test.

(a) $H_1 : \mu \neq \mu_0$, $\alpha = 0.10$
(b) $H_1 : \mu < \mu_0$, $\alpha = 0.01$
(c) $H_1 : p > p_0$, $\alpha = 0.05$

1.4 A provincial office tests whether a job-training programme raised the mean monthly income of participants. H_0: the programme has no effect on mean income; H_1: the programme raises mean income. Describe a Type I error and a Type II error in this context, and give one consequence of each.

1.5 The unemployment rate in a province is 5% (hypothetical figure). Can the normal approximation be used for $\hat{p}$ when (a) $n = 80$, (b) $n = 200$? Explain.

1.6 A 95% CI for the mean monthly electricity bill of households in a city is 1,120 to 1,280 baht. Which statements are correct?

(a) We are 95% confident that the mean monthly electricity bill of all households in the city is between 1,120 and 1,280 baht.
(b) 95% of households in the city pay between 1,120 and 1,280 baht per month.
(c) If we took many samples and built a 95% CI from each, about 95% of the intervals would contain the population mean.

Solutions

Solution: Your Turn 1.1

(a) $\sigma_{\bar{x}} = 900/\sqrt{81} = 100$ baht.

(b) $z = (4320 - 4500)/100 = -1.80$, so $P(\bar{x} < 4320) = P(Z > 1.80) = 1 - 0.9641 = 0.0359$.

Solution: Your Turn 1.2

$np = 240$ and $n(1-p) = 360$ are both at least 5. $\sigma_{\hat{p}} = \sqrt{0.4(0.6)/600} = 0.0200$, $z = (0.45 - 0.4)/0.0200 = 2.50$, so $P(\hat{p} > 0.45) = 1 - 0.9938 = 0.0062$.

Solution: Your Turn 1.3

(a) $H_0 : \mu = 200000 \quad H_1 : \mu > 200000$ (right-sided)
(b) $H_0 : p = 0.50 \quad H_1 : p \neq 0.50$ (two-sided)
(c) $H_0 : \mu = 14 \quad H_1 : \mu < 14$ (left-sided)

Solution: Your Turn 1.4

(a) Areas 0.005 and 0.995: $z_{0.995} = 2.576$. Critical region: $z_{\text{cal}} < -2.576$ or $z_{\text{cal}} > 2.576$.

(b) Area 0.99: $z_{0.99} = 2.326$. Critical region: $z_{\text{cal}} > 2.326$.

(c) Area 0.10: $z_{0.10} = -z_{0.90} = -1.282$. Critical region: $z_{\text{cal}} < -1.282$.

Solution: Exercise 1.1

(a) $\sigma_{\bar{x}} = 4/\sqrt{64} = 0.5$ baht.

(b) $z = (44.2 - 45)/0.5 = -1.60$; $P(\bar{x} < 44.2) = 1 - 0.9452 = 0.0548$.

(c) $z = (46 - 45)/0.5 = 2.00$; $P(44.2 < \bar{x} < 46) = 0.9772 - 0.0548 = 0.9224$.

Solution: Exercise 1.2

(a) $H_0 : \mu = 3 \quad H_1 : \mu > 3$
(b) $H_0 : p = 0.70 \quad H_1 : p \neq 0.70$
(c) $H_0 : \mu = 20 \quad H_1 : \mu < 20$

Solution: Exercise 1.3

(a) $z_{0.95} = 1.645$: reject H_0 if $z_{\text{cal}} < -1.645$ or $z_{\text{cal}} > 1.645$.
(b) $z_{0.01} = -z_{0.99} = -2.326$: reject H_0 if $z_{\text{cal}} < -2.326$.
(c) $z_{0.95} = 1.645$: reject H_0 if $z_{\text{cal}} > 1.645$.

Solution: Exercise 1.4

Type I error: concluding that the programme raises income when it does not. Budget may continue to be spent on a programme that does not work. Type II error: concluding that the programme has no effect when it does raise income. A useful programme may be cut.

Solution: Exercise 1.5

(a) $np = 80(0.05) = 4 < 5$, so the normal approximation should not be used.
(b) $np = 200(0.05) = 10$ and $n(1-p) = 190$ are both at least 5, so it can be used.

Solution: Exercise 1.6

(a) and (c) are correct. (b) is wrong: the interval is about the population mean, not about individual households.

2 One-Sample Inference

2.1 Mean, σ Known

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 2.1.1

Problem to be added.

Step 1: Hypotheses

H_0: _____

H_1: _____

Step 2: Level of significance

$\alpha =$ _____

Step 3: Test statistic

Formula: _____

Substitution: _____

Result: _____

Step 4: Critical value and critical region

Critical value(s): _____

Critical region: _____

Step 5: Decision and interpretation

Decision: _____

_____

Interpretation: _____

_____

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 2.1.1

Solution to be added.

2.2 Mean, σ Unknown

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 2.2.1

Problem to be added.

Step 1: Hypotheses

H_0: _____

H_1: _____

Step 2: Level of significance

$\alpha =$ _____

Step 3: Test statistic

Formula: _____

Substitution: _____

Result: _____

Step 4: Critical value and critical region

Critical value(s): _____

Critical region: _____

Step 5: Decision and interpretation

Decision: _____

_____

Interpretation: _____

_____

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 2.2.1

Solution to be added.

2.3 Proportion

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 2.3.1

Problem to be added.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 2.3.1

Solution to be added.

2.4 Variance

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 2.4.1

Problem to be added.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 2.4.1

Solution to be added.

3 Two-Sample Inference

3.1 Two Proportions

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.1.1

Problem to be added.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.1.1

Solution to be added.

3.2 P-values and R Output

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.2.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.2.1

Solution to be added.

3.3 Paired Means

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.3.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.3.1

Solution to be added.

3.4 Two Variances

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.4.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.4.1

Solution to be added.

3.5 Two Independent Means

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.5.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.5.1

Solution to be added.

4 Analysis of Variance

4.1 One-Way ANOVA: CRD

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 4.1.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 4.1.1

Solution to be added.

4.2 Multiple Comparisons (Tukey)

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 4.2.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 4.2.1

Solution to be added.

4.3 Randomized Complete Block Design (RCBD)

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 4.3.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 4.3.1

Solution to be added.

Summary: Choosing a Method

Which Method? A Flowchart

To be added.

Mixed Problems

To be added.

Writing a Full Conclusion

To be added.